

Subject Description Form

Subject Code	COMP5423
Subject Title	Natural Language Processing
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil (but some knowledge in artificial intelligence is preferable)
Objectives	The objectives of this subject are to enable students to acquire: 1. essential knowledge of natural language processing and its application scenarios; 2. practical techniques to handle and process natural language text; 3. fundamental skills to develop systems with natural language processing techniques and tools; and 4. basic building blocks to design innovative software for different communities.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand and apply fundamental knowledge and advanced techniques of natural language processing for textual information processing; b. develop creative solutions to solve technical problems through critical thinking, analytical thinking and creative thinking, leveraging emerging technologies for global challenges; c. demonstrate the ability to design, implement and evaluate effective natural language processing systems and applications to fulfill the real-world needs, demonstrating critical awareness and integrity in making responsible decisions; and d. engage in life-long independent learning for professional development, continuously adapting and reflecting on personal growth.
Subject Synopsis/ Indicative Syllabus	Natural Language Processing Core Concepts Morphological Analysis, Syntactic Analysis and Parsing, Semantic Analysis, Discourse Analysis, and the Problem of Ambiguity. Natural Language Processing Models and Techniques Text Segmentation, Classification, Ranking, Sequence Labeling, N-Gram Language Model, Topic Model, Role of Machine Learning and Data Mining in Natural Language Processing, Natural Language Processing Tools, and Large Language Models (LLMs). Natural Language Processing Applications Machine Translation, Question Answering, Summarization, Opinion Mining and Sentiment Classification, Dialogue and Conversation, Reading Comprehension, Information Retrieval and Extraction, News Recommendation, etc. Selected Topics in Natural Language Processing with Deep Learning Word Embedding (Word2Vec), Convolutional Neural Network, Recurrent Neural Network, Recursive Neural Network, Sequence to Sequence Model.

Teaching/Learning Methodology	Lectures teach students on the main concepts of the course, together with comprehensive examples, and class questions and answers for easy understanding. Tutorials and lab sessions help students to review the learned concepts, master the practical techniques and necessary tools for effective system/application development. Group project offers the opportunity to students to develop analytical and problem solving skills through system implementation and interpersonal communication.																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Assignment, and Project</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Final Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Lecture assignment and final examination evaluate student’s ability to understand fundamental knowledge, to analyze and solve the problem problems through critical thinking, analytical thinking and creative thinking. Lab assignment and project evaluate student’s ability to apply learned techniques and tools to design and implement real-world application systems. Project also evaluates their abilities of independent learning and group communication.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Assignment, and Project	30	✓	✓	✓	✓	2. Final Examination	70	✓	✓	✓		Total	100				
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Reading List and References	Reference Books: (1) Daniel Jurafsky and James H. Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition. Prentice Hall. 2000. (2) Christopher D. Manning and Hinrich Schütze. Foundations of Statistical Natural Language Processing. The MIT Press. 1999. (3) Nitin Indurkha and Fred J. Damerau. Handbook of Natural Language Processing. Chapman and Hall/CRC. 2010. (4) Steven Bird, Ewan Klein and Edward Loper. Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O’Reilly Media. 2009.																																	

	(5) Julia Silge and David Robinson. Text Mining with R. O'Reilly Media. 2018. (6) Li Deng and Yang Liu, Deep Learning in Natural Language Processing. Springer. 2018. (7) Yoav Goldberg and Graeme Hirst. Neural Network Methods for Natural Language Processing. Morgan & Claypool Publishers. 2017. (8) Jason Brownlee. Deep Learning for Natural Language Processing. Machine Learning Mastery, 2018.
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